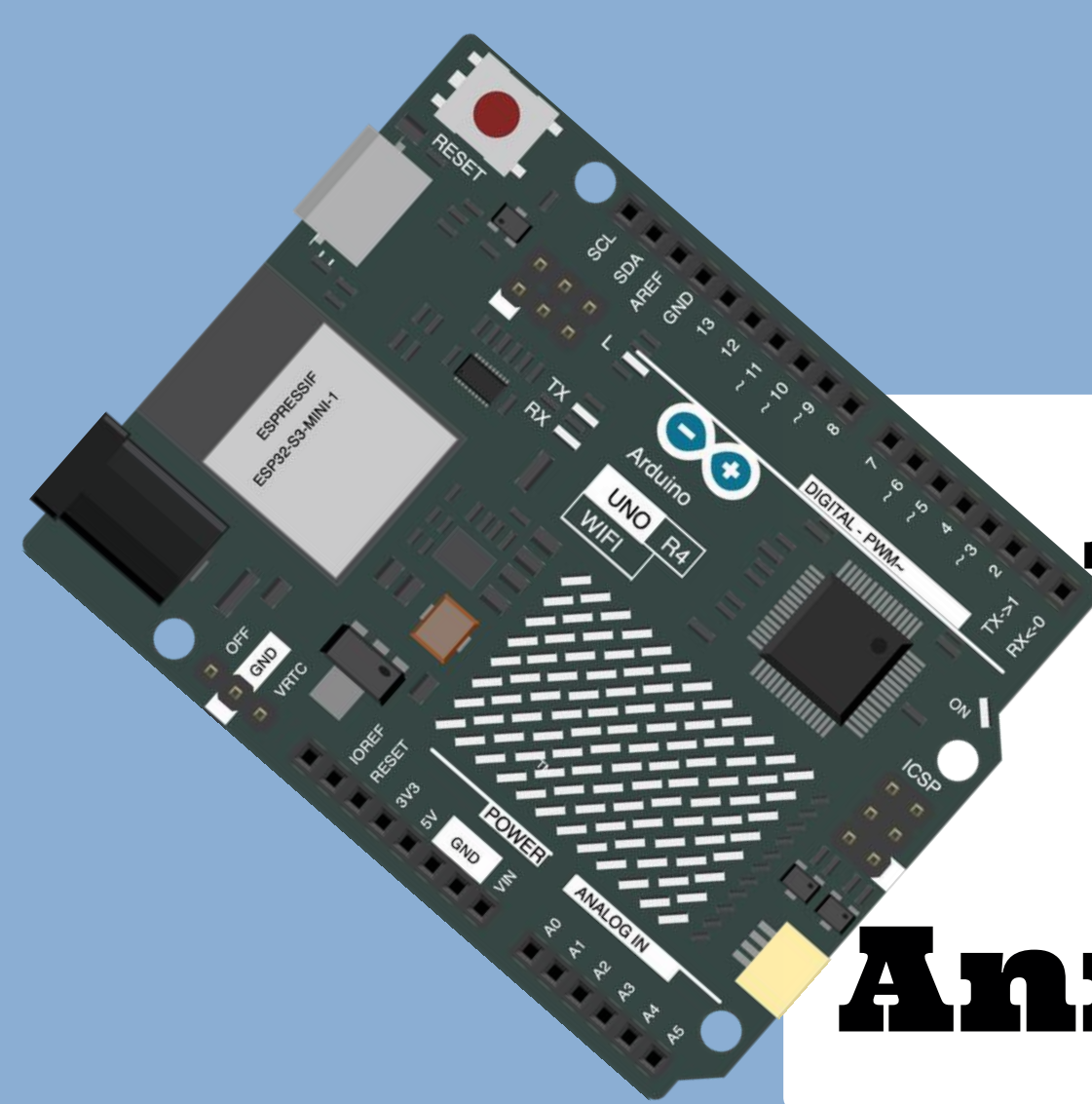
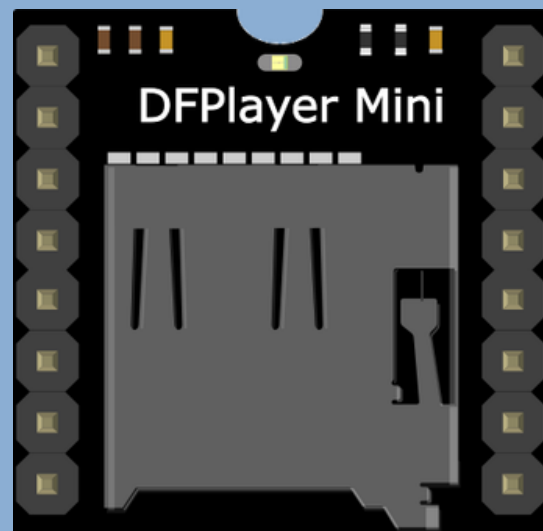




Automatic School Buzzer and Announcement System

Using Arduino Uno R4, DF Player Mini, Speaker, and Webpage
School Name: Abhinav Vidya Mandir, VIDYA Tech Fest 2026-

27



Problem Statement

Manual bell ringing causes delays and disruptions.

Announcements need constant staff presence.

Schools urgently need automation solutions.

Time and effort are wasted on manual tasks.

By the end of this project, you'll be able to...



Ring the school bell
automatically



Play recorded
announcements



Show current time on
Dashboard Webpage

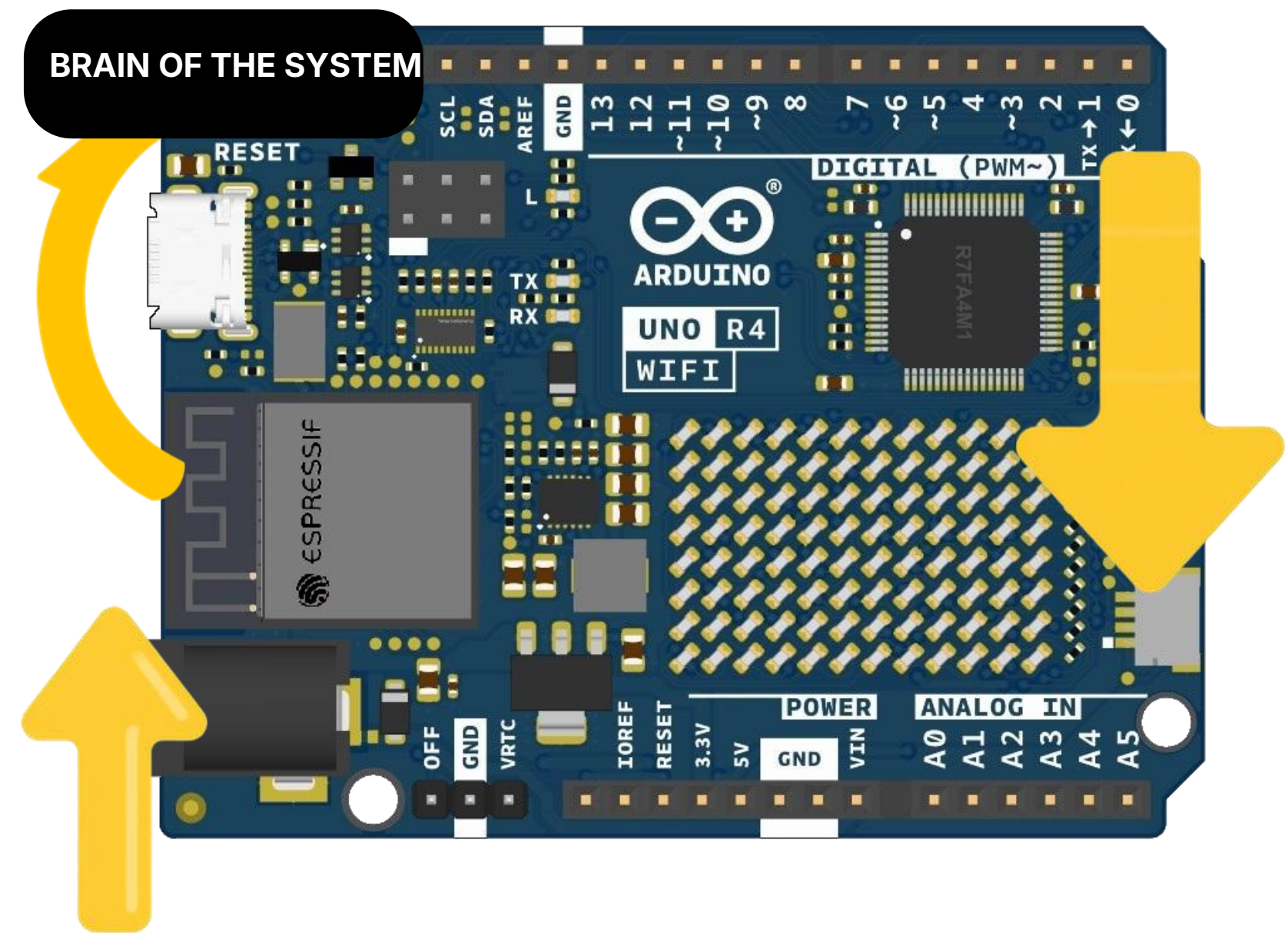


Reduce manual work in
schools

Introduction to Arduino

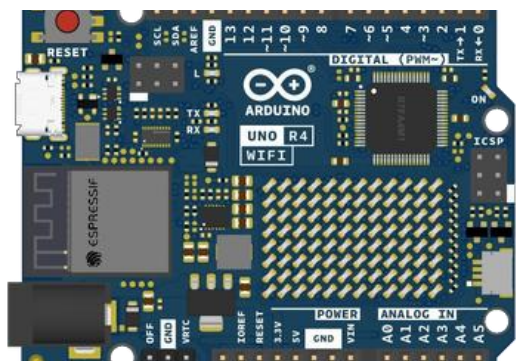
Arduino R4 Wifi is a microcontroller board that controls electronic devices and components.

It is easy to use in school projects and is the main controller of this system.



Components Required

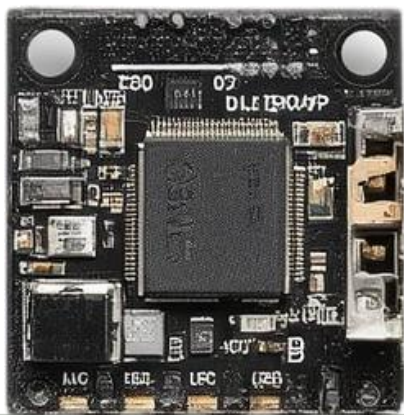
Below are the key components used in the Automatic School Buzzer and Announcement System:



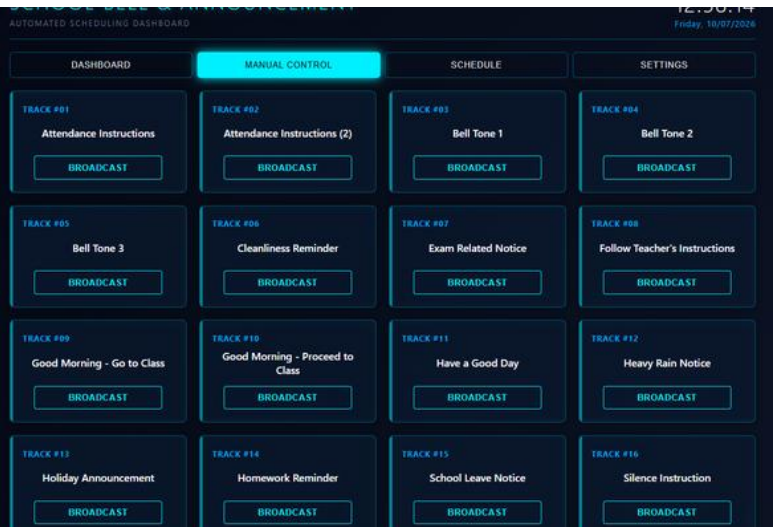
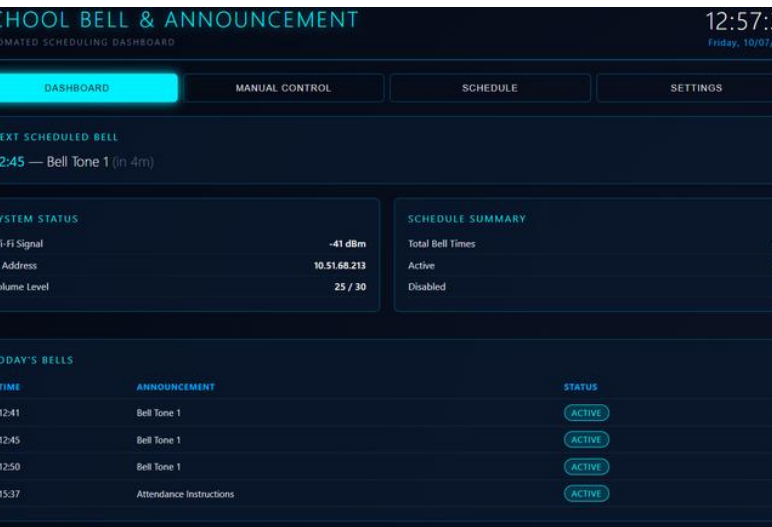
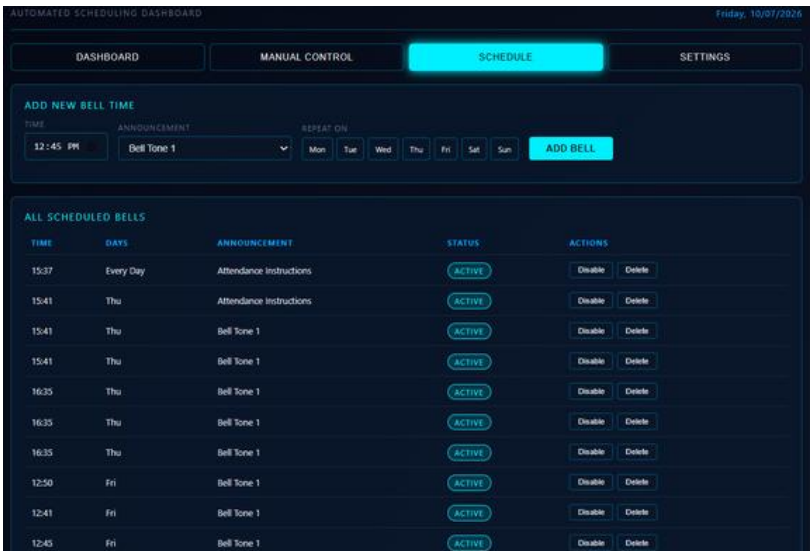
Arduino R4 WIFI



Speaker



DF Player Mini



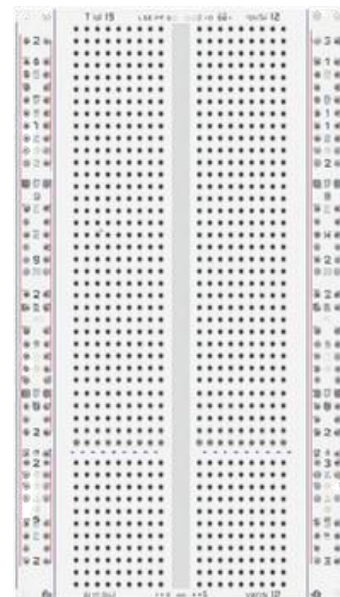
Website
Dashboard

Components Required (cont.)

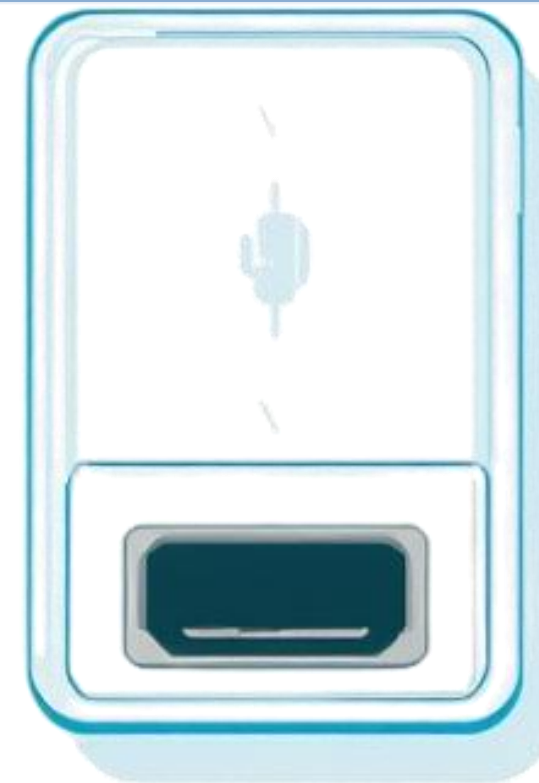
Additional parts needed
to build the circuit.



Jumper Wires
Connect components
on the breadboard
easily.



Breadboard
Prototype
circuits without
soldering.

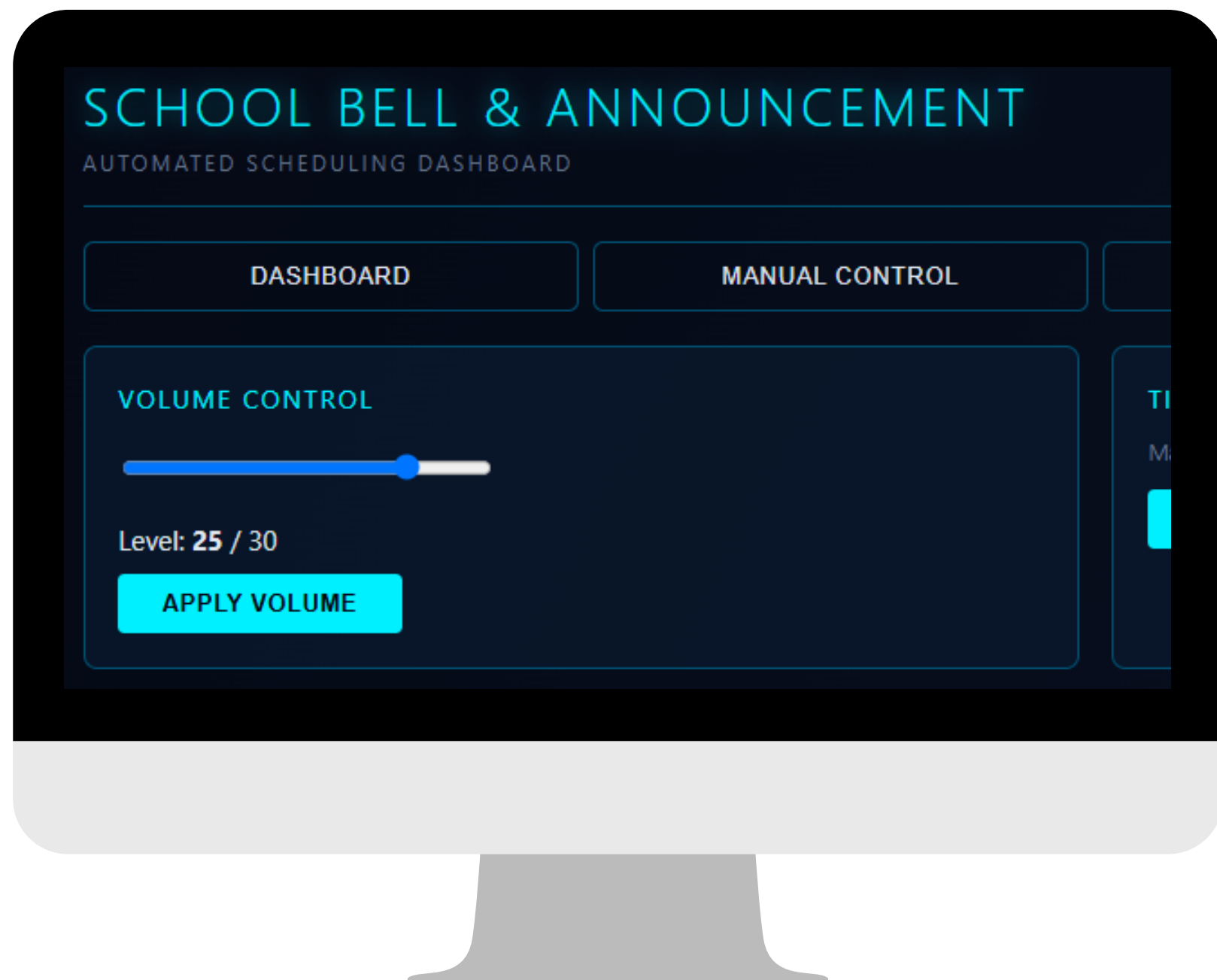


Power Supply
Provides stable power to
Arduino and modules.



Micro SD Card
Stores audio
announcement files for
DF Player Mini.

Website Dashboard



Arduino Uno R4 Wifi → Webpage Display
(Time input to controller to screen output)



Arduino Uno R4 Wifi → DF Player Mini → Speaker
(Controller triggers audio playback)



Arduino Uno → Buzzer
(Controller rings the bell at scheduled times)

Circuit Connection Diagram & Code

WIRING GUIDE

```
School_Bell_Systemv2.ino
1  /*
2
3  SCHOOL BELL & ANNOUNCEMENT CONTROL SYSTEM - Arduino UNO R4 WiFi + DFPlayer
4  */
5
6  Features:
7  - Professional tabbed web dashboard (Dashboard / Manual / Schedule / Settings)
8  - NTP time sync using the onboard RTC (RA4M1) so time survives resets
9  (keeps ticking even if WiFi drops, re-syncs automatically once a day)
10 - Bell/announcement SCHEDULER: add/edit/delete/enable/disable time slots,
11 each tied to a specific day-of-week pattern and a specific audio track
12 - Persistent storage of the schedule in the UNO R4's emulated EEPROM
13 (survives power loss / reset)
14 - Manual "Broadcast Now" grid for all 16 announcements
15 - Volume control + manual re-sync button
16
17 REQUIRED LIBRARIES (Arduino IDE > Library Manager):
18 - DFRobotDFPlayerMini (by DFRobot)
19 - NTPClient (by Fabrice Weinberg / arduino-libraries)
20 "RTC.h" and "WiFiS3.h" ship automatically with the "Arduino UNO R4 Boards" core.
21
22 IMPORTANT - SD CARD FILE NAMING:
23 The DFPlayer Mini's playMp3Folder() call requires files inside a folder
24 named "mp3" on the SD card, named with 4-digit numbers:
25 /mp3/0001.mp3
26 ...
27 /mp3/0016.mp3
28 Your uploaded files must be renamed into that scheme. Based on the files
29 you gave me, I mapped them in upload order to tracks 1-16 below. If your
30 SD card is in a different order, just re-order the ANNOUNCEMENT_NAMES
31 array (and rename the physical files on the SD card) to match - nothing
32 else in the code needs to change.
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Audio Announcement Feature

Morning Assembly Announcement
Recess Bell Announcement
School Closing Announcement
Emergency Message Playback

Advantages

Key benefits of the Automatic School Buzzer and Announcement System:



Fully Automatic



Saves Teacher's Time



Accurate Bell Timing



Clear Announcements



Easy to Modify



Low Cost

Real-Life Applications & Future Improvements

Applications

- ✓ Schools
- ✓ Colleges
- ✓ Factories
- ✓ Hospitals
- ✓ Offices
- ✓ Railways

Colleges

Schools

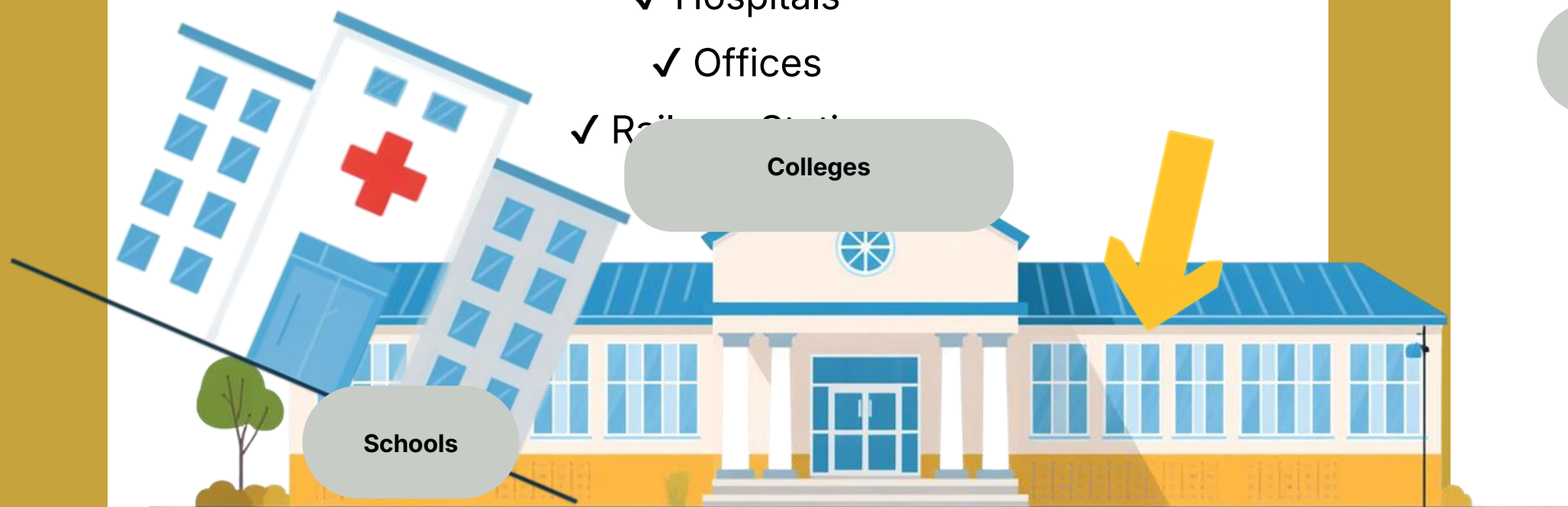
Future Ideas

- ◆ Mobile App Control
- ◆ Wi-Fi Monitoring
- ◆ Multiple Bell Schedules
- ◆ Voice Recording
- ◆ Cloud-Based Management

Hospitals

Factories

Offices



**Thank
You!**

